

SYNOPSIS

Register No: 127004158

Name: MOHAMED IBRAHIM K

Project Title: Prediction of Crop Yield Based-on Soil Parameters using Machine Learning Algorithms

Name of the Guide: Dr. ARUN K, Assistant Professor III, SEEE, SASTRA DEEMED UNIVERSITY, TANJORE.

Abstract

Modern Agriculture uses soil sensors and weather data to suggest what to plant. This project builds a crop-recommendation system that reads soil and local data in real time.

A 7-in-1 soil sensor measures nitrogen (N), phosphorus (P), potassium (K), and pH. Temperature, humidity, and rainfall come from weather APIs. An ESP32 sends the readings to a Raspberry Pi gateway, which runs an ensemble model built from boosting methods and displays the best recommended crop.

We tested the system across different input sets and it produced stable recommendations. It can help farmers choose crops, raise yields, and reduce wasted water and fertilizer.

Specific Contribution

- Designed the IoT data gathering setup using a 7-in-1 soil sensor and an ESP32.
- Built an ensemble crop-prediction model using boosting methods.
- Integrated the hardware, cloud layer, and ML model so the system runs end to end without manual steps.

Specific Learning

- Explored how to combine soil and weather inputs to make crop decisions.
- Built hands-on skill with machine learning, especially boosting-based ensemble models.
- Worked through data preparation, model training, and end-to-end system integration issues.

Technical Limitations & Ethical Challenges faced

- The system depends on stable internet access for weather APIs and cloud updates.
- A small dataset can limit how well the model holds up across new farms, seasons, and regions.
- Keeping recommendations consistent across soil types and locations and avoiding biased outputs was harder than expected.

Keywords: Smart Agriculture, Crop Recommendation, IoT, Machine Learning, Ensemble Models, Soil Sensors, Precision Farming



Signature of the Student

Date:06/05/2026



Signature of Guide